CODE SPORT

Sandya Mannarswamy

In this month's column, we begin a discussion on 'data science' and also feature a set of systems software interview questions.

Let's continue with another set of interview questions for systems software engineers. But before we get to the interview practice questions, I wanted to update you on what I plan to discuss in our column over the next few months. Many of our readers have written to me with requests to discuss topics from one of the emerging areas of information technology, namely data science.

Data science is being seen as the next big wave in the software industry. Unlike the developments in computer architecture or operating systems, this is a change that is going to have a direct impact not just in enterprise business technologies, but also in the way information is consumed by individuals. So I wanted to start off this month's column with a brief introduction to data science. Before we go into what it is, let us talk about why data has become the most significant commodity in information technology.

We produce 2.5 quintillion bytes of information every day. This information is generated by our Web searches, the online purchases we make, our mobile calls and our social network presence (a quintillion is 1000 X 1000 X 1000 times a billion). Given the volume, velocity, variety and variability of data that gets produced, expertise is needed to make sense out of the vast quantities of data, in order to derive meaningful, actionable information from it. Indeed, this has given rise to a new breed of computer scientists known as 'data scientists'. They are a rare breed since data science itself is a discipline in its infancy. In fact, a McKinsey report from May 2011 predicts a huge shortage of data scientists, estimating that in the US alone, more than 190,000 data scientists would be needed.

Even today, we frequently encounter many 'data science' applications in our daily interactions. The customer reviews and product recommendations you see on Amazon are an example of the use of data science. Customer reviews are opinions mined to identify and compare different features of the product under discussion and presented in a concise way to the consumer. Mining opinion reviews for positive and negative sentiments involve different elements of data science such as natural language processing, machine learning and analytics. Product recommendations are based on 'recommender systems', which use the profile data associated with customers to identify potential products that they may be interested in. These data science applications deal mostly with unstructured data, unlike traditional enterprise or e-commerce applications, which typically deal with structured data housed in databases.

The main difference between traditional data applications involving databases and the newer breed of data science applications is the role of unstructured data, and the fact that multiple data sources are combined, mined and analysed to get actionable information that can be used to drive other applications. As O'Reilly puts it, "Data is the next Intel inside." Given the vast volumes of data around us, there are infinite possibilities for analysing and processing them. For instance, during the swine flu epidemic of 2009, Google could build a visual trail of the disease's spread across the different states of the US by correlating the search history and frequency of users 'googling' for swine flu related topics.

'Data science' requires multi-disciplinary skills in a number of areas. These include probability, statistics, information retrieval, Web search, analytics, natural language processing, data mining and machine learning. It's only now that universities are waking up to offering courses on data science. There are not many books that provide a comprehensive overview of the subject. Given the interest expressed by many of our readers and its importance to budding